

FBISE MATHEMATICS HSSC-I

1st Annual Examination 2026

HARD AREA PAPER

Complete, Exam-Ready Solutions

All 20 MCQs with reasoning and option analysis

All parts of Sections B and C, including every OR alternative

Step-by-step working and required diagrams

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Educational solution guide based on the supplied question paper.

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How to use these solutions

Write the relevant formula first, substitute clearly, show intermediate steps, and box the final answer with units where applicable. Both alternatives are solved here for study; in the examination, attempt only the number of alternatives required by the paper.

Quick answer key – Section A

Q	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	A	A	D	C	D	A	C

Q	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	B	A	D	D	A	C	D

1 Section A - MCQs with Full Reasoning

MCQ 1: Factorize $z^2 + 169$.

Since $169 = 13^2$ and $i^2 = -1$,

$$z^2 + 169 = z^2 - (13i)^2 = (z + 13i)(z - 13i).$$

Answer: B, $(z + 13i)(z - 13i)$.

Why the other options are wrong: Option A gives $z^2 - 169$. Option C repeats the same factor and gives $(z - 13i)^2$, while D gives $(z + 13i)^2$; both contain a nonzero middle term.

MCQ 2: Find the argument of $z = 1 + i$.

z is in the first quadrant and $\tan \theta = \frac{1}{1} = 1$, so $\theta = \pi/4$.

Answer: B, $\pi/4$. **Why the other options are wrong:** $\pi/6$ and $\pi/3$ have tangents $1/\sqrt{3}$ and $\sqrt{3}$; $\pi/2$ would require zero real part.

MCQ 3: Rank of an identity matrix.

Every row and column of an $n \times n$ identity matrix contains a pivot. Hence its rank is n , equal to its order.

Answer: B, equal to its order. **Why the other options are wrong:** It is neither smaller nor greater than its order; rank is well-defined.

MCQ 4: A system is inconsistent if it has...

An inconsistent system contains contradictory equations and therefore has no common solution.

Answer: A, no solution. **Why the other options are wrong:** A unique solution or infinitely many solutions makes a system consistent. "Repeated solutions" is not the defining condition.

MCQ 5: Fifth term of an A.P. with $a_1 = 2$, $d = 3$.

$$a_5 = a_1 + (5 - 1)d = 2 + 4(3) = 14.$$

Answer: A, 14. **Why the other options are wrong:** 11 is a_4 , 13 does not follow the common difference, and 15 would arise from an incorrect starting value.

MCQ 6: If $a_1, a_2, a_3, \dots$ is an A.P., its corresponding H.P. is...

An H.P. is a sequence whose reciprocals form an A.P. Thus it is

$$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$$

Answer: D. **Why the other options are wrong:** The original terms remain an A.P.; their products, sums, or the single reciprocal of their sum do not define the corresponding H.P.

MCQ 7: Coefficient of the middle term.

In a binomial expansion, the middle term and its coefficient depend on the exponent n ; for example, the relevant coefficient is a binomial coefficient such as $\binom{n}{n/2}$ when n is even.

Answer: C, depends on n . **Why the other options are wrong:** It is not always 1 or 0, and it is not independent of n .

MCQ 8: First step in mathematical induction.

One first verifies the proposition for the initial value, usually $n = 1$. This is the base case.

Answer: D, base case. **Why the other options are wrong:** The induction hypothesis comes afterward; the result/conclusion is reached only after the inductive step.

MCQ 9: Degree of remainder when a cubic is divided by a linear polynomial.

The remainder must have degree smaller than the divisor. A linear divisor has degree 1, so the remainder is constant and has degree 0 (or it is the zero polynomial).

Answer: A, 0. Why the other options are wrong: Degrees 1, 2, 3 are not smaller than the divisor's degree.

MCQ 10: Factor theorem: if $f(a) = 0$, then...

By the factor theorem, $f(a) = 0$ exactly when $(x - a)$ is a factor of $f(x)$.

Answer: C, $(x - a)$ is a factor. Why the other options are wrong: A number a itself is a zero, not a polynomial factor; $(x + a)$ would correspond to $f(-a) = 0$; $f(a)$ equals zero and is not stated as a factor.

MCQ 11: Two nonzero vectors have zero dot product.

Since $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = 0$ and both magnitudes are nonzero, $\cos \theta = 0$, so $\theta = \pi/2$.

Answer: C, mutually perpendicular. Why the other options are wrong: Parallel/equal vectors have angle 0 and a positive dot product; opposite vectors have angle π and a negative dot product.

MCQ 12: Dot product in component form.

For $\vec{a} = a_1\hat{i} + b_1\hat{j}$ and $\vec{b} = a_2\hat{i} + b_2\hat{j}$,

$$\vec{a} \cdot \vec{b} = a_1a_2 + b_1b_2.$$

Answer: A. Why the other options are wrong: Option B has the wrong sign; C mixes unlike components; D is the two-dimensional determinant, not the dot product.

MCQ 13: Angle between orthogonal vectors.

Orthogonal vectors meet at a right angle, $\theta = 90^\circ = \pi/2$.

Answer: B, $\pi/2$. Why the other options are wrong: π is opposite direction, 2π is the same direction, and $2\pi/3$ is 120° .

MCQ 14: Evaluate $\cos(\pi + \beta)$.

$$\cos(\pi + \beta) = \cos \pi \cos \beta - \sin \pi \sin \beta = -\cos \beta.$$

Answer: B, $-\cos \beta$. Why the other options are wrong: The remaining options have either the wrong sign or the wrong trigonometric function.

MCQ 15: Product-to-sum form of $\cos A \cos B$.

Adding $\cos(A + B)$ and $\cos(A - B)$ gives $2 \cos A \cos B$. Hence

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)].$$

Answer: A. Why the other options are wrong: B and C omit one required term and the factor $1/2$; D is not the cosine product identity.

MCQ 16: Period of $\cos(2x/5)$.

For $\cos(kx)$, $T = 2\pi/|k|$. Here $k = 2/5$, so

$$T = \frac{2\pi}{2/5} = 5\pi.$$

Answer: D, 5π . Why the other options are wrong: 2π is the basic cosine period before horizontal scaling; $5\pi/2$ and $2\pi/5$ use the scale factor incorrectly.

MCQ 17: Which function is even?

f is even if $f(-x) = f(x)$. Since $\cos(-x) = \cos x$, cosine is even.

Answer: D, $f(x) = \cos x$. Why the other options are wrong: x^3 , $\sin x$, and x all satisfy $f(-x) = -f(x)$ and are odd.

MCQ 18: Maximum of $y = 3 + 2 \sin \theta$.

Because $-1 \leq \sin \theta \leq 1$, the maximum occurs at $\sin \theta = 1$:

$$y_{\max} = 3 + 2(1) = 5.$$

Answer: A, 5. Why the other options are wrong: 4, 3, and 2 are below the attainable maximum.

MCQ 19: Arrangements of the letters of GARDEN.

The six letters are distinct, so the number of arrangements is $6! = 720$.

Answer: C, 720. Why the other options are wrong: 480, 600, and 840 are not $6!$; no division for repeated letters is needed.

MCQ 20: Choose 10 members from 15.

Order does not matter, so

$$\binom{15}{10} = \binom{15}{5} = \frac{15!}{10!5!} = 3003.$$

Answer: D, 3003. Why the other options are wrong: $360,360$, $15!/10!$, and $15!/5!$ count ordered selections or omit a factorial from the combination formula.

2 Section B - Short Questions (All Alternatives)

Q.2(i) Solve $z + 2\omega = 4i$, $2z + 5\omega = 3$.

Multiply the first equation by 2:

$$2z + 4\omega = 8i.$$

Subtract it from the second equation:

$$\omega = 3 - 8i.$$

Then

$$z = 4i - 2(3 - 8i) = -6 + 20i.$$

$$\boxed{z = -6 + 20i, \quad \omega = 3 - 8i}$$

Check: $z + 2\omega = (-6 + 20i) + (6 - 16i) = 4i$.

OR

A force $\vec{F} = 2\hat{i} + 3\hat{j} + \hat{k}$ **causes displacement** $\vec{d} = \hat{i} + 2\hat{j} + 2\hat{k}$. **Find work.**

$$W = \vec{F} \cdot \vec{d} = (2)(1) + (3)(2) + (1)(2) = 10.$$

$$\boxed{W = 10 \text{ J}}$$

Q.2(ii) Verify $\sin 4\theta = 8 \sin \theta \cos^3 \theta - 4 \sin \theta \cos \theta$.

Starting from the right side,

$$\begin{aligned} 8 \sin \theta \cos^3 \theta - 4 \sin \theta \cos \theta &= 4 \sin \theta \cos \theta (2 \cos^2 \theta - 1) \\ &= 2 \sin 2\theta \cos 2\theta \\ &= \sin 4\theta. \end{aligned}$$

Hence LHS = RHS. Verified

OR

Evaluate $\frac{(n-1)!}{(n-3)!}$.

$$\frac{(n-1)!}{(n-3)!} = \frac{(n-1)(n-2)(n-3)!}{(n-3)!} = \boxed{(n-1)(n-2)}, \quad n \geq 3.$$

Q.2(iii) Volume of tetrahedron formed by $\vec{a} = 2\hat{i} + \hat{j}$, $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$, $\vec{c} = \hat{j} + \hat{k}$.

The volume is one-sixth of the absolute scalar triple product:

Handwritten calculation:

$$V = \frac{1}{6} \begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{vmatrix} = \frac{1}{6} (2 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 0 + 0 \cdot 1 \cdot 3 - 0 \cdot 1 \cdot 1 - 1 \cdot 1 \cdot 2 - 2 \cdot 1 \cdot 1) = \frac{1}{6} (2 - 2 - 2) = \frac{1}{6} (-2) = -\frac{1}{3}$$

Final result:

$$V = \frac{1}{6} |3| = \frac{1}{2} \text{ cubic unit}$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 2\hat{i} - \hat{j} + \hat{k}.$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (2, 1, 0) \cdot (2, -1, 1) = 4 - 1 = 3.$$

$$\boxed{V = \frac{1}{6} |3| = \frac{1}{2} \text{ cubic unit}}$$

OR

Find maximum and minimum values of $y = \frac{1}{2 + 3 \sin(2x + 1)}$.

Since $-1 \leq \sin(2x + 1) \leq 1$, the denominator ranges from -1 to 5 . However, it can also equal zero when $\sin(2x + 1) = -2/3$. At those points the function is undefined and tends to $\pm\infty$. Therefore there is no finite global maximum or minimum.

More precisely,

$$\text{Range}(y) = (-\infty, -1] \cup [1/5, \infty).$$

The least positive value is $1/5$ (when $\sin(2x + 1) = 1$), and the greatest negative value is -1 (when $\sin(2x + 1) = -1$). This distinction is essential because the denominator crosses zero.

Q.2(iv) Find K if $x - 1$ is a factor of $f(x) = x^3 + K^2 - 2K + 1$.

By the factor theorem, $f(1) = 0$:

$$1 + K^2 - 2K + 1 = 0 \implies K^2 - 2K + 2 = 0.$$

$$K = \frac{2 \pm \sqrt{4 - 8}}{2} = 1 \pm i.$$

$$K = 1 + i \text{ or } K = 1 - i$$

There is no real value of K , since $K^2 - 2K + 2 = (K - 1)^2 + 1 > 0$ for real K .

OR

Find the sum of six terms of $1 + \frac{1}{2} + \frac{1}{4} + \dots$. This is a G.P. with $a = 1$, $r = 1/2$, $n = 6$:

$$S_6 = \frac{a(1 - r^6)}{1 - r} = \frac{1 - (1/2)^6}{1 - 1/2} = \frac{63}{32}.$$

Q.2(v) For $y = 2 \sin(3x + 1)$, determine amplitude, period, and range.

Comparing with $y = A \sin(Bx + C)$,

$$\text{Amplitude} = |A| = 2,$$

$$\text{Period} = \frac{2\pi}{|B|} = \frac{2\pi}{3},$$

$$-2 \leq y \leq 2.$$

OR

Find the angle between $\vec{a} = 4\hat{i} + \hat{j} - 8\hat{k}$ **and** $\vec{b} = \hat{i} + 4\hat{j} + \hat{k}$.

$$\vec{a} \cdot \vec{b} = 4(1) + 1(4) + (-8)(1) = 0.$$

Both vectors are nonzero, hence $\cos \theta = 0$ and

$$\theta = \frac{\pi}{2} = 90^\circ.$$

Q.2(vi) Find the 10th term of an A.P. with $a_1 = 3$ and $a_7 = 15$.

$$a_7 = a_1 + 6d \implies 15 = 3 + 6d \implies d = 2.$$

$$a_{10} = a_1 + 9d = 3 + 18 = 21.$$

OR

Using the fundamental law of trigonometry, find $\cos 15^\circ$.

$$\begin{aligned}\cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \frac{1}{2} = \boxed{\frac{\sqrt{6} + \sqrt{2}}{4}}.\end{aligned}$$

Q.2(vii) Find the rank of $A = \begin{bmatrix} 3 & -4 & 2 \\ 2 & 3 & 5 \\ 1 & 0 & 1 \end{bmatrix}$.

$$|A| = 3(3) - (-4)(2 - 5) + 2(0 - 3) = 9 - 12 - 6 = -9 \neq 0.$$

A nonzero determinant means all three rows/columns are independent, so

$$\boxed{\text{rank}(A) = 3}.$$

OR

Express $2 \sin 12\theta \cos 6\theta$ as a sum or difference. Using $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$,

$$\boxed{2 \sin 12\theta \cos 6\theta = \sin 18\theta + \sin 6\theta}.$$

Q.2(viii) A G.P. has $a_3 = 16$ and $a_5 = 64$. Find r , a_1 , and the positive-value sequence.

$$ar^2 = 16, \quad ar^4 = 64.$$

Dividing, $r^2 = 4$, so $r = \pm 2$. In either case $a = 16/r^2 = 4$.

$$\boxed{r = 2, a = 4: \quad 4, 8, 16, 32, 64, \dots}$$

The other algebraic possibility is $r = -2$, giving $4, -8, 16, -32, 64, \dots$; the requested positive-value sequence uses $r = 2$.

OR

Find the inverse of $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 2 \\ 2 & 1 & 4 \end{bmatrix}$.

$$|A| = 24 \neq 0, \quad \text{adj } A = \begin{bmatrix} 10 & -4 & 2 \\ 4 & 8 & -4 \\ -6 & 0 & 6 \end{bmatrix}.$$

Therefore

$$A^{-1} = \frac{1}{24} \begin{bmatrix} 10 & -4 & 2 \\ 4 & 8 & -4 \\ -6 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 5/12 & -1/6 & 1/12 \\ 1/6 & 1/3 & -1/6 \\ -1/4 & 0 & 1/4 \end{bmatrix}.$$

Q.2(ix) Determine the last two digits of 13^{10} .

Work modulo 100:

$$\begin{aligned}13^2 &= 169 \equiv 69, & 13^4 &\equiv 69^2 = 4761 \equiv 61, \\ 13^8 &\equiv 61^2 = 3721 \equiv 21.\end{aligned}$$

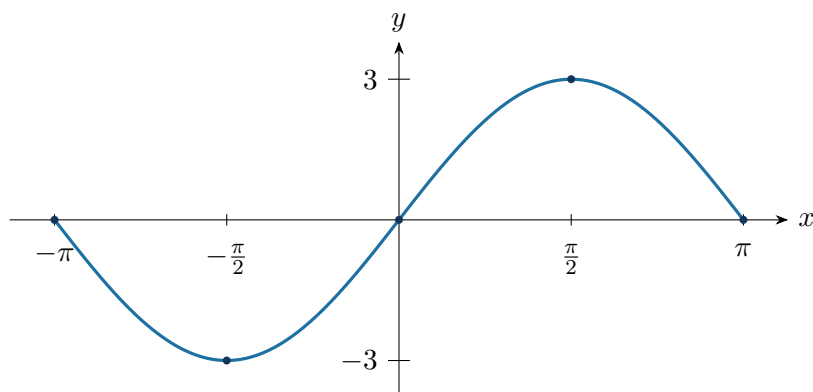
Thus

$$13^{10} = 13^8 \cdot 13^2 \equiv 21 \cdot 69 = 1449 \equiv \boxed{49} \pmod{100}.$$

Hence the last two digits are 49.

OR

Sketch $y = 3 \sin x$ **for** $-\pi \leq x \leq \pi$. Key points are $(-\pi, 0)$, $(-\pi/2, -3)$, $(0, 0)$, $(\pi/2, 3)$, $(\pi, 0)$.



Q.2(x) Convert $z = 1 + i$ **to polar form and find** z^2 .

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \theta = \tan^{-1}(1) = \frac{\pi}{4}.$$

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right).$$

By De Moivre's theorem,

$$z^2 = 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = \boxed{2i}.$$

OR

Find r **if** ${}^{54}P_r : {}^{56}P_{r+3} = 1 : 30800$.

$$\begin{aligned} \frac{{}^{54}P_r}{{}^{56}P_{r+3}} &= \frac{54!}{(54-r)!} \cdot \frac{(53-r)!}{56!} \\ &= \frac{1}{56 \cdot 55 \cdot (54-r)} = \frac{1}{30800}. \end{aligned}$$

Thus $56 \cdot 55(54-r) = 30800$. Since $56 \cdot 55 = 3080$, $54-r = 10$, and

$$\boxed{r = 44}.$$

Q.2(xi) Use the Binomial Theorem to find the remainder when 8^{205} **is divided by** 48.

Write $8^2 = 64 = 48 + 16$:

$$8^{205} = 8(8^2)^{102} = 8(48 + 16)^{102}.$$

In the binomial expansion, every term containing a positive power of 48 is divisible by 48. Therefore

$$8^{205} \equiv 8(16^{102}) \pmod{48}.$$

Also $16^2 = 256 = 5(48) + 16$, and hence repeatedly $16^m \equiv 16 \pmod{48}$ for every $m \geq 1$. Thus

$$8^{205} \equiv 8(16) = 128 \equiv \boxed{32} \pmod{48}.$$

The required remainder is 32.

OR

Factorize $x^3 - 4x^2 + x + 6$. Testing integral roots, $f(2) = 8 - 16 + 2 + 6 = 0$, so $(x - 2)$ is a factor. Division gives

$$x^3 - 4x^2 + x + 6 = (x - 2)(x^2 - 2x - 3).$$

Since $x^2 - 2x - 3 = (x - 3)(x + 1)$,

$$x^3 - 4x^2 + x + 6 = (x - 2)(x - 3)(x + 1).$$

Q.2(xii) Find λ if $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & \lambda \\ 3 & 1 & 2 \end{bmatrix}$ is singular.

For a singular matrix, the determinant is zero:

$$\begin{aligned} 0 &= 1(2 - \lambda) - (-1)(4 - 3\lambda) + 2(2 - 3) \\ &= 2 - \lambda + 4 - 3\lambda - 2 = 4 - 4\lambda. \end{aligned}$$

Therefore $\lambda = 1$.

OR

Use induction to prove $1 + 2 + \dots + n = \frac{n(n+1)}{2}$.

Base case $n = 1$: LHS = 1 and RHS = $1(2)/2 = 1$; true.

Induction hypothesis: Assume for $n = k$ that

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}.$$

Inductive step: For $n = k + 1$,

$$\begin{aligned} 1 + 2 + \dots + k + (k + 1) &= \frac{k(k+1)}{2} + (k + 1) \\ &= (k + 1) \left(\frac{k}{2} + 1 \right) \\ &= \frac{(k + 1)(k + 2)}{2}. \end{aligned}$$

This is the required form with $n = k + 1$. Hence the statement is true for all $n \in \mathbb{N}$.

3 Section C - Long Questions (All Alternatives)

Q.3: Profit polynomial $P(x) = x^3 + 6x^2 - x - 6$; **find the 2012 profit using division by** $x - 2$.

Since x is years after 2010, the year 2012 corresponds to $x = 2$. By the remainder theorem, division by $x - 2$ leaves remainder $P(2)$:

$$P(2) = 2^3 + 6(2^2) - 2 - 6 = 8 + 24 - 2 - 6 = \boxed{24}.$$

Thus the modeled profit for 2012 is 24 in the model's stated profit units.

Verification by long division:

$$\begin{array}{r|rrrr} x-2 & x^3 & +6x^2 & -x & -6 \\ & \underline{x^3} & \underline{-2x^2} & & \\ & & 8x^2 & -x & \\ & & \underline{8x^2} & \underline{-16x} & \\ & & & 15x & -6 \\ & & & \underline{15x} & \underline{-30} \\ & & & & 24 \end{array}$$

Hence

$$P(x) = (x - 2)(x^2 + 8x + 15) + 24,$$

confirming remainder 24.

OR

Let $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ **and** $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$.

(a) Find $\vec{a} \times \vec{b}$:

$$\begin{aligned} \vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 4 \\ 1 & 2 & -1 \end{vmatrix} \\ &= \hat{i}(2 - 8) - \hat{j}(-3 - 4) + \hat{k}(6 + 2) \\ &= \boxed{-6\hat{i} + 7\hat{j} + 8\hat{k}}. \end{aligned}$$

(b) Verify $\vec{a} \times \vec{b} \perp \vec{a}$:

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = (-6)(3) + 7(-2) + 8(4) = -18 - 14 + 32 = 0.$$

Therefore $\vec{a} \times \vec{b} \perp \vec{a}$.

(c) Verify $\vec{a} \times \vec{b} \perp \vec{b}$:

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = (-6)(1) + 7(2) + 8(-1) = -6 + 14 - 8 = 0.$$

Therefore $\vec{a} \times \vec{b} \perp \vec{b}$.

Q.4: Find the term independent of x in $(2x - 1/x)^{12}$ and also the 10th term.

The general term is

$$\begin{aligned} T_{r+1} &= \binom{12}{r} (2x)^{12-r} \left(-\frac{1}{x}\right)^r \\ &= \binom{12}{r} 2^{12-r} (-1)^r x^{12-2r}. \end{aligned}$$

Independent term: Set the exponent of x equal to zero:

$$12 - 2r = 0 \implies r = 6.$$

Thus the independent term is the 7th term:

$$T_7 = \binom{12}{6} 2^6 = 924(64) = \boxed{59136}.$$

10th term: $r + 1 = 10 \implies r = 9$:

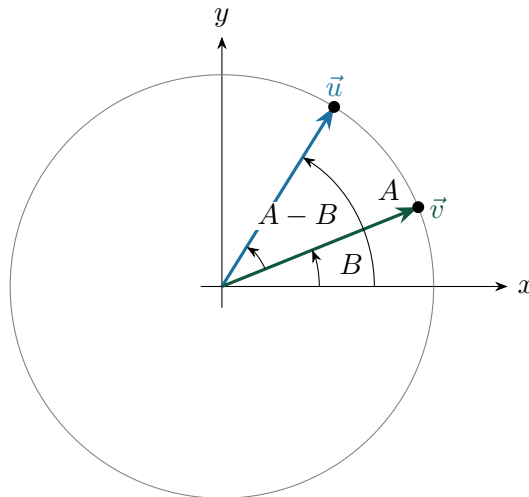
$$T_{10} = \binom{12}{9} 2^3 (-1)^9 x^{12-18} = -220(8)x^{-6} = \boxed{-\frac{1760}{x^6}}.$$

OR

Prove $\cos(A - B) = \cos A \cos B + \sin A \sin B$. **Diagram included.**

Take two unit vectors $\vec{u}$ and $\vec{v}$ making angles A and B with the positive x -axis. Their coordinates are

$$\vec{u} = (\cos A, \sin A), \quad \vec{v} = (\cos B, \sin B).$$



The angle between the unit vectors is $A - B$. Using the geometric dot-product formula,

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos(A - B) = \cos(A - B),$$

because $|\vec{u}| = |\vec{v}| = 1$. Using components,

$$\vec{u} \cdot \vec{v} = \cos A \cos B + \sin A \sin B.$$

Equating the two expressions gives

$$\boxed{\cos(A - B) = \cos A \cos B + \sin A \sin B}.$$

Q.5: Solve $x + y + z = 6$, $2x + 3y + z = 10$, $x + 2y + 3z = 11$.

Label the equations (1), (2), and (3). Subtract (1) from (2):

$$x + 2y = 4. \tag{4}$$

Using (4) in (3),

$$4 + 3z = 11 \implies z = \frac{7}{3}.$$

Equation (1) gives

$$x + y = 6 - \frac{7}{3} = \frac{11}{3}. \quad (5)$$

Subtract (5) from (4):

$$y = 4 - \frac{11}{3} = \frac{1}{3}.$$

Then $x = \frac{11}{3} - \frac{1}{3} = \frac{10}{3}$. Therefore

$$\boxed{x = \frac{10}{3}, \quad y = \frac{1}{3}, \quad z = \frac{7}{3}}.$$

Substitution in all three original equations verifies the solution.

OR

Ferris wheel: $h(t) = 10 + 8 \sin\left(\frac{\pi}{6}t\right)$.

(a) Period. For $\sin(\omega t)$, $T = 2\pi/\omega$:

$$\boxed{T = \frac{2\pi}{\pi/6} = 12 \text{ time units}}.$$

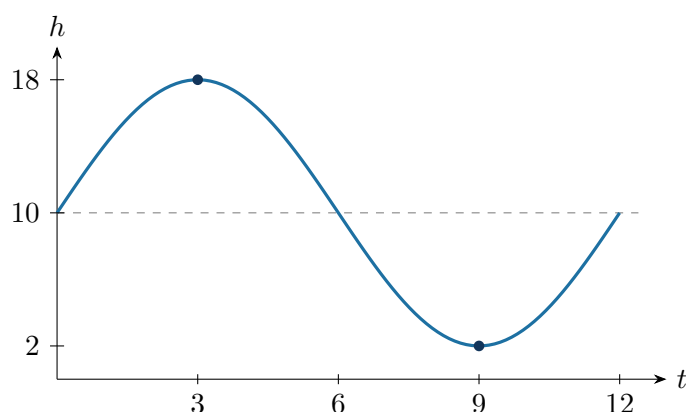
(b) Maximum and minimum height. Since $-1 \leq \sin(\pi t/6) \leq 1$,

$$\boxed{h_{\max} = 10 + 8 = 18}, \quad \boxed{h_{\min} = 10 - 8 = 2}.$$

(c) Time of maximum height. The maximum occurs when the sine equals 1:

$$\frac{\pi}{6}t = \frac{\pi}{2} + 2\pi k \implies \boxed{t = 3 + 12k}, \quad k \in \mathbb{Z}.$$

For $t \geq 0$, the first maximum occurs at $\boxed{t = 3}$.



Q.6(a) For 2, 5, 8, 11, ..., find the general term and sum of first 20 terms.

Here $a = 2$ and $d = 3$. Therefore

$$a_n = a + (n - 1)d = 2 + 3(n - 1) = \boxed{3n - 1}.$$

The 20th term is $a_{20} = 3(20) - 1 = 59$. Hence

$$S_{20} = \frac{20}{2}(a_1 + a_{20}) = 10(2 + 59) = \boxed{610}.$$

Q.6(b) Find the sum of the first n terms of $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots$. The k th term is $k(k+1) = k^2 + k$. Thus

$$\begin{aligned} S_n &= \sum_{k=1}^n (k^2 + k) \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \\ &= \frac{n(n+1)}{6} [(2n+1) + 3] \\ &= \boxed{\frac{n(n+1)(n+2)}{3}}. \end{aligned}$$

OR

A box contains 7 red, 5 blue, and 4 green balls. In how many ways can 5 balls be selected if at least 2 are red?

There are 9 non-red balls. Count by the number r of selected red balls:

$$\begin{aligned} N &= \binom{7}{2} \binom{9}{3} + \binom{7}{3} \binom{9}{2} + \binom{7}{4} \binom{9}{1} + \binom{7}{5} \binom{9}{0} \\ &= 21(84) + 35(36) + 35(9) + 21(1) \\ &= 1764 + 1260 + 315 + 21 \\ &= \boxed{3360 \text{ ways}}. \end{aligned}$$

Final Examination Checklist

- State the formula before substitution, especially for A.P./G.P., permutations, vectors, and binomial terms.
- In vector questions, show component calculations and the zero dot product used to prove perpendicularity.
- For the sine graphs, mark axes, scale, amplitude, zeros, maxima, minima, and the stated interval.
- In induction, explicitly write the base case, hypothesis, $k + 1$ step, and conclusion.
- For a singular matrix, set the determinant equal to zero. For an inverse, first confirm the determinant is nonzero.
- In the reciprocal-trigonometric question, note that the denominator can be zero; therefore there is no global maximum or minimum.

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